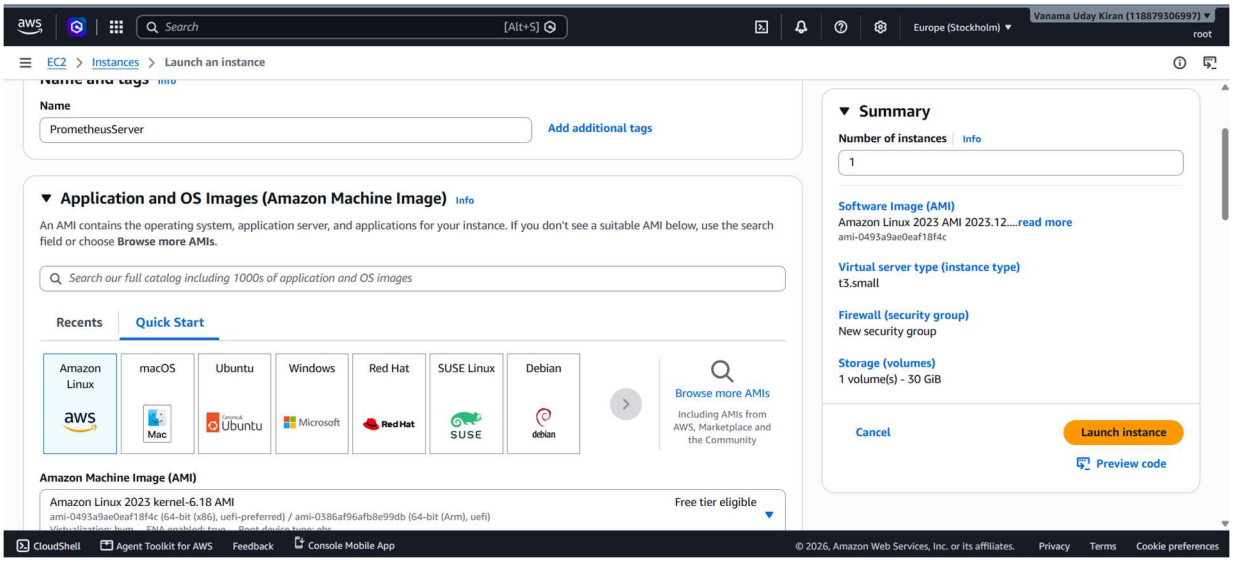
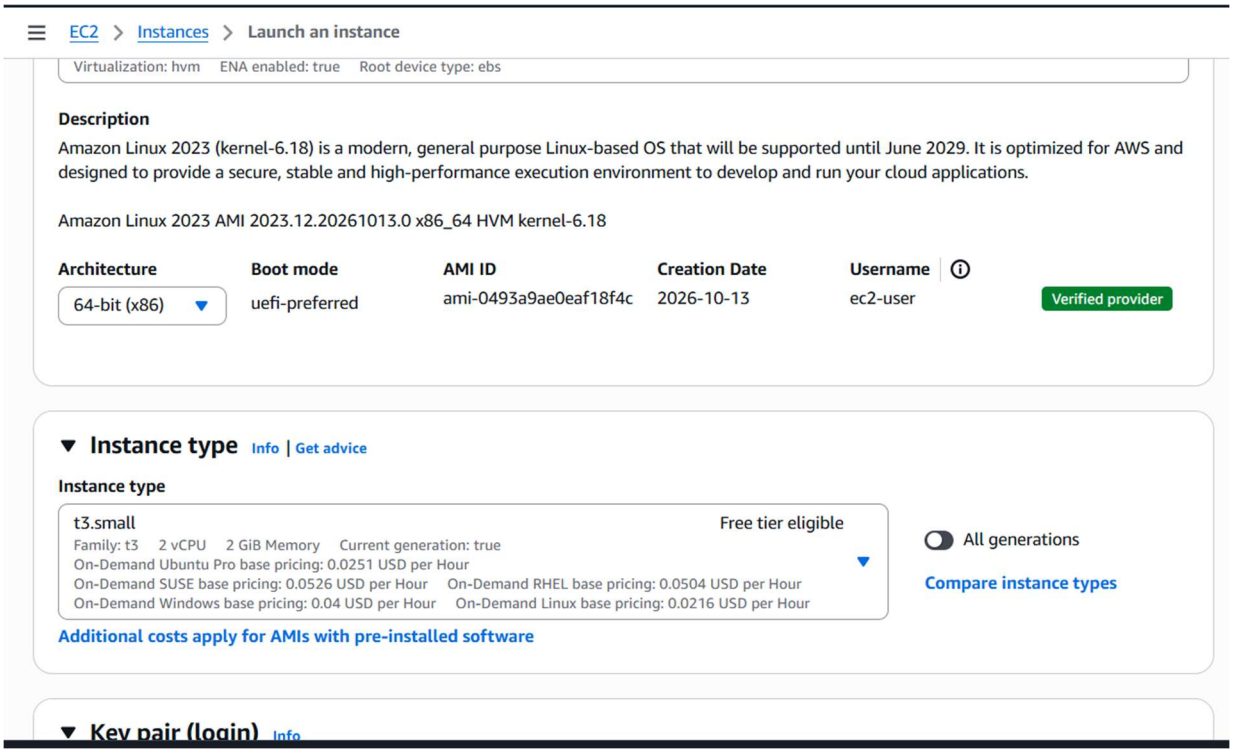
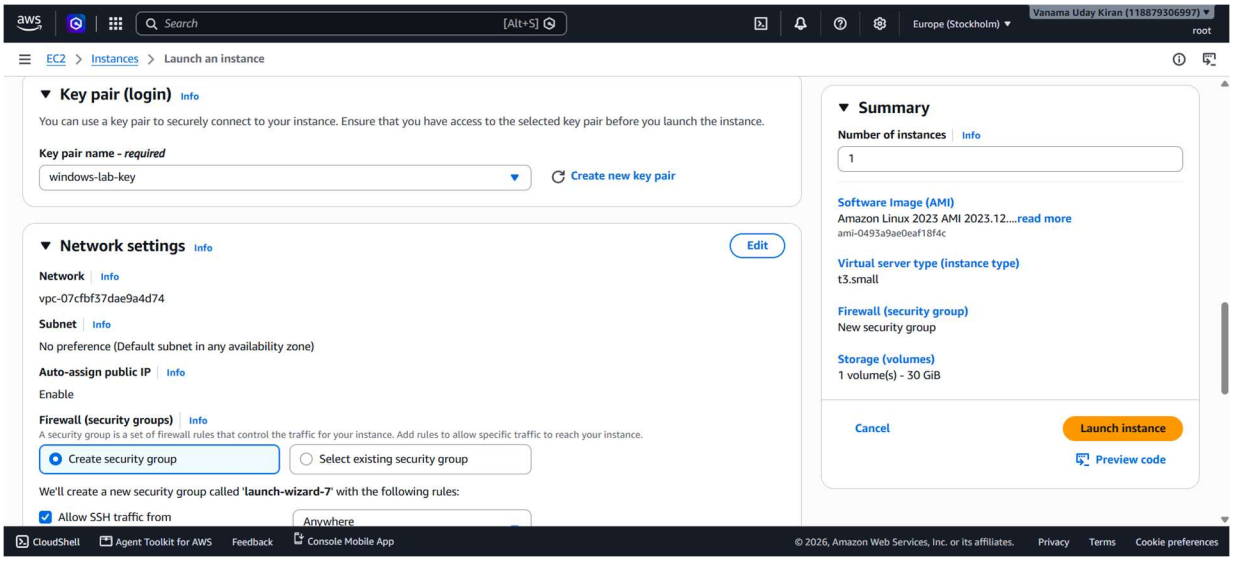
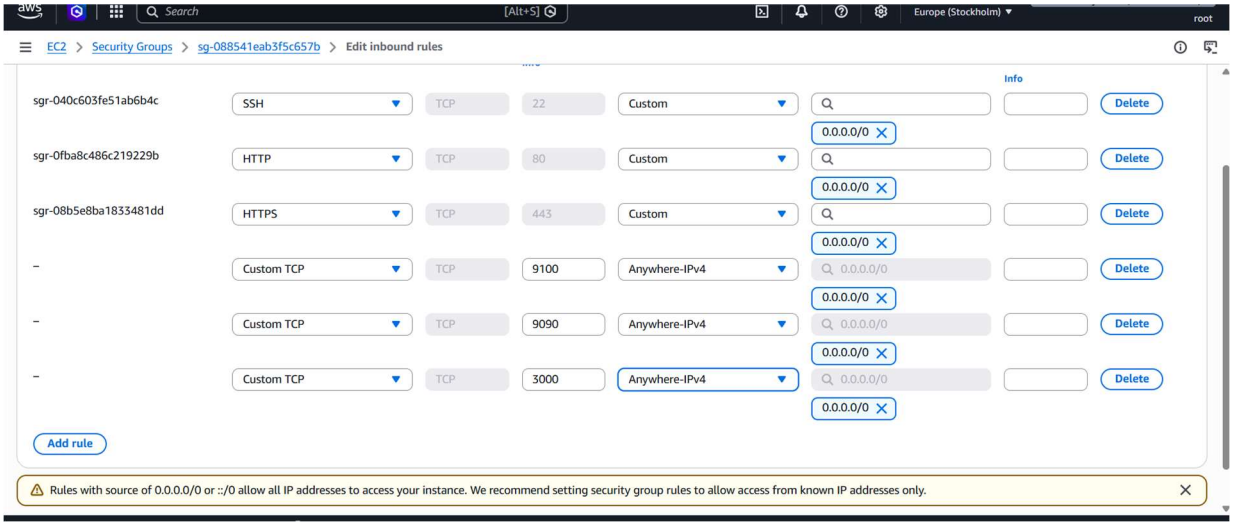
Create a Linux Main server Instance



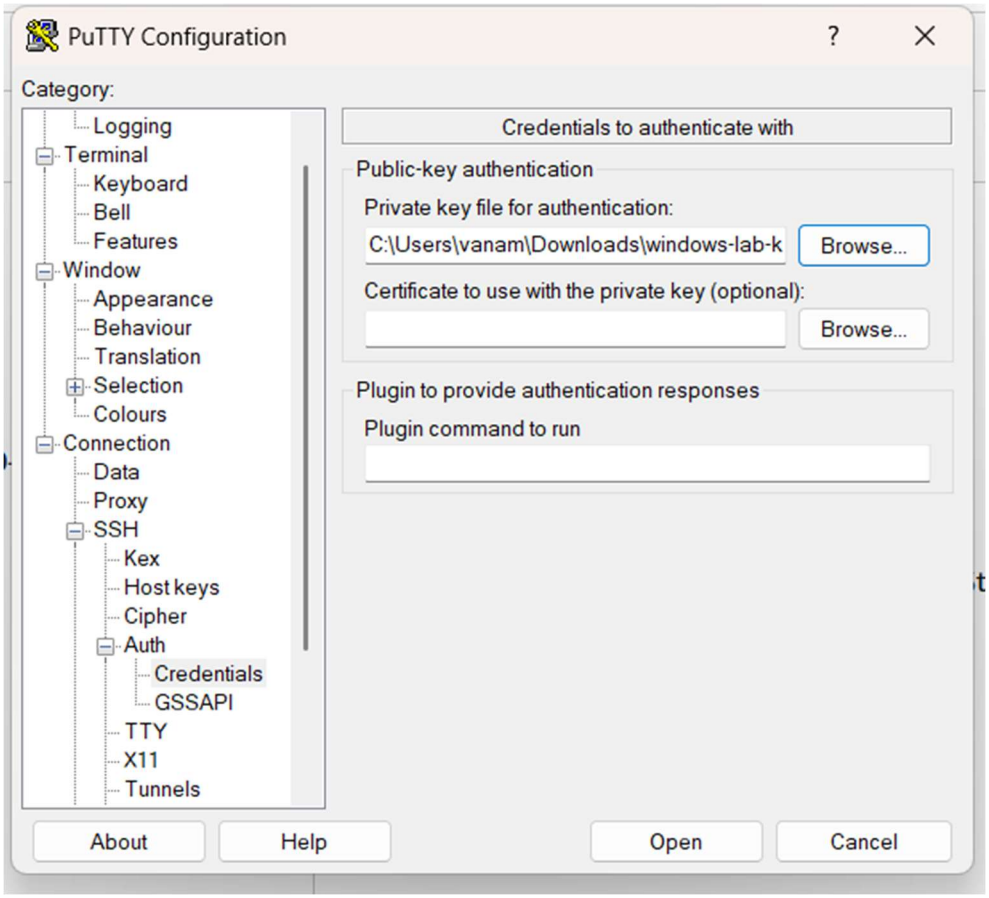




Add the security rules:



Connecting Putty


PuTTY Configuration
Category:
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credentials
GSSAPI
TTY
X11
Tunnels
Credentials to authenticate with
Public-key authentication
Private key file for authentication:
C:\Users\vanam\Downloads\windows-lab-k
Browse...
Certificate to use with the private key (optional):
Browse...
Plugin to provide authentication responses
Plugin command to run
About
Help
Open
Cancel


login as: ec2-user
Authenticating with public key "imported-openssh-key"
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-1-184 ~]$ sudo su -
[root@ip-172-31-1-184 ~]# wget https://github.com/prometheus/prometheus/releas
/download/v3.15.0/prometheus-3.15.0.linux-amd64.tar.gz

```
11651578/111651578]

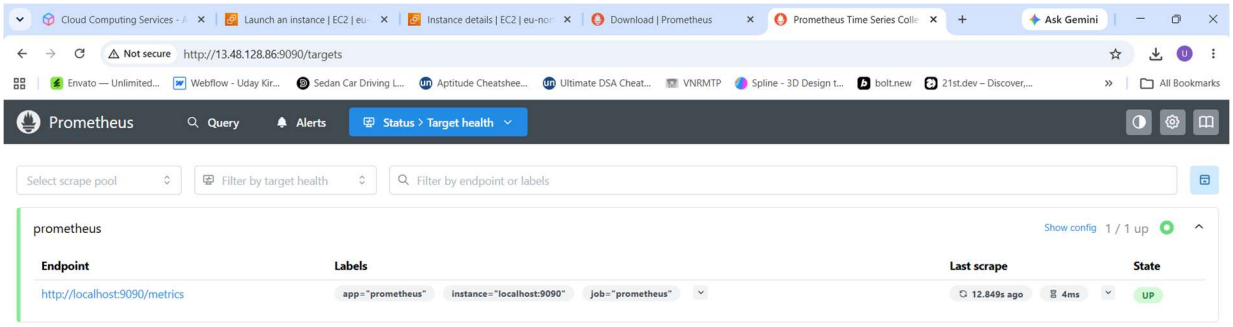
[root@ip-172-31-1-184 ~]# ls
prometheus-3.15.0.linux-amd64.tar.gz
[root@ip-172-31-1-184 ~]# tar -xvzf prometheus-3.15.0.linux-amd64.tar.gz
prometheus-3.15.0.linux-amd64/
prometheus-3.15.0.linux-amd64/promtool
```

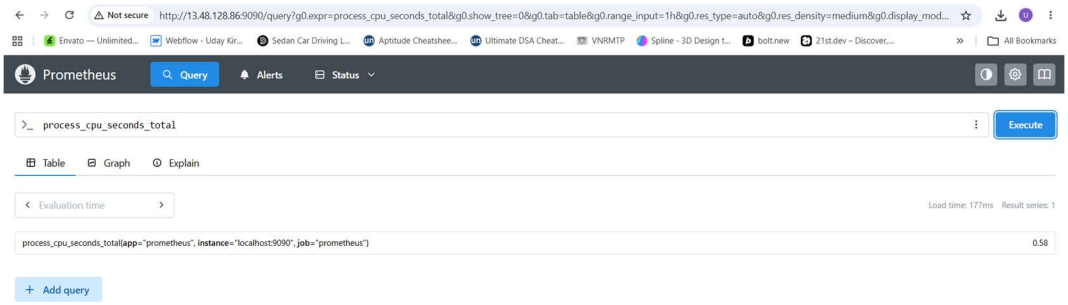
```
prometheus-3.15.0.linux-amd64/prometheus.yml
[root@ip-172-31-1-184 ~]# ls
prometheus-3.15.0.linux-amd64  prometheus-3.15.0.linux-amd64.tar.gz
[root@ip-172-31-1-184 ~]# cd
.ssh/                          prometheus-3.15.0.linux-amd64/
[root@ip-172-31-1-184 ~]# cd
.ssh/                          prometheus-3.15.0.linux-amd64/
[root@ip-172-31-1-184 ~]# cd
.ssh/                          prometheus-3.15.0.linux-amd64/
[root@ip-172-31-1-184 ~]# cd prometheus-3.15.0.linux-amd64/
[root@ip-172-31-1-184 prometheus-3.15.0.linux-amd64]# ls
LICENSE  NOTICE  prometheus  prometheus.yml  promtool
[root@ip-172-31-1-184 prometheus-3.15.0.linux-amd64]# cp prometheus /usr/local/bin
```

```
[root@ip-172-31-1-184 ~]# cd prometheus-3.15.0.linux-amd64/
[root@ip-172-31-1-184 prometheus-3.15.0.linux-amd64]# ls
LICENSE  NOTICE  prometheus  prometheus.yml  promtool
[root@ip-172-31-1-184 prometheus-3.15.0.linux-amd64]# cp prometheus /usr/local/bin
[root@ip-172-31-1-184 prometheus-3.15.0.linux-amd64]# cp promtool /usr/local/bin
[root@ip-172-31-1-184 prometheus-3.15.0.linux-amd64]# prometheus --version
prometheus, version 3.15.0 (branch: HEAD, revision: 5241a27fe3c6983549fccc32f6e65917408c63cd)
  build user:       root@4635dd007a60
  build date:       20260925-07:33:45
  go version:       go1.27.1
  platform:         linux/amd64
  tags:             netgo,builtinassets
[root@ip-172-31-1-184 prometheus-3.15.0.linux-amd64]#
```

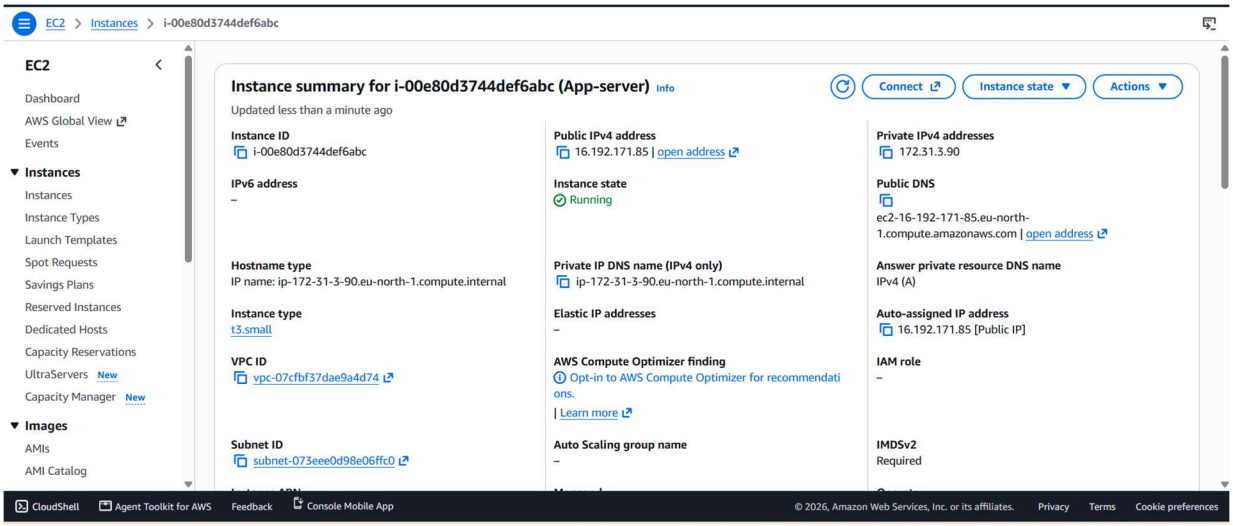
```
root@ip-172-31-1-184 prometheus-3.15.0.linux-amd64]# prometheus --version
prometheus, version 3.15.0 (branch: HEAD, revision: 5241a27fe3c6983549fccc32f6e65917408c63cd)
  build user:       root@4635dd007a60
  build date:       20260925-07:33:45
  go version:       go1.27.1
  platform:         linux/amd64
  tags:             netgo,builtinassets
root@ip-172-31-1-184 prometheus-3.15.0.linux-amd64]# ./prometheus --config.file=prometheus.yml
ime=2026-10-15T11:00:18.417Z level=INFO source=main.go:1770 msg="updated GOGC" old=100 new=75
ime=2026-10-15T11:00:18.418Z level=INFO source=main.go:824 msg="Leaving GOMAXPROCS=2: CPU quota undefi
 component=automaxprocs
ime=2026-10-15T11:00:18.419Z level=INFO source=memlimit.go:198 msg="GOMEMLIMIT is updated" component=a
emlimit package=github.com/KimMachineGun/automemlimit/memlimit GOMEMLIMIT=1802115072 previous=92233720
4775807
ime=2026-10-15T11:00:18.419Z level=INFO source=main.go:933 msg="Starting Prometheus Server" mode=serve
rsion="(version=3.15.0, branch=HEAD, revision=5241a27fe3c6983549fccc32f6e65917408c63cd)"
ime=2026-10-15T11:00:18.419Z level=INFO source=main.go:938 msg="operational information" build_contexta
```

Checking





Creating the EC2 Node Exporter :

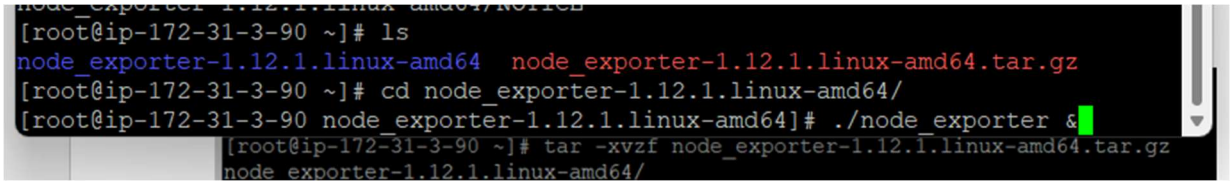




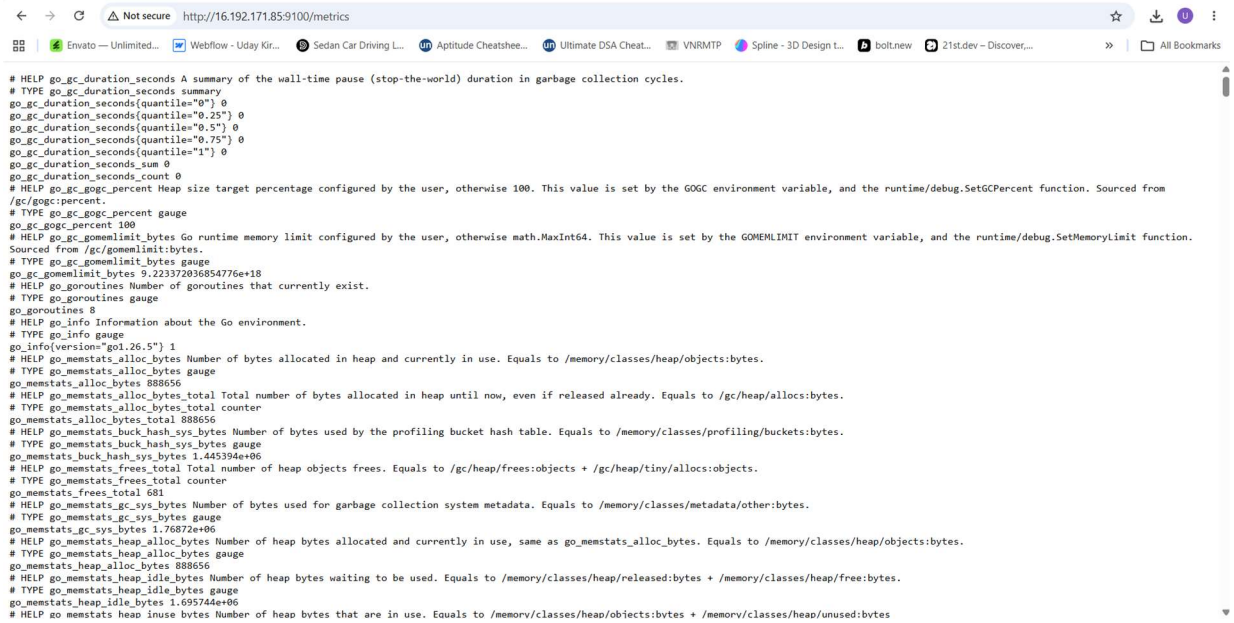
Installing node exporter :



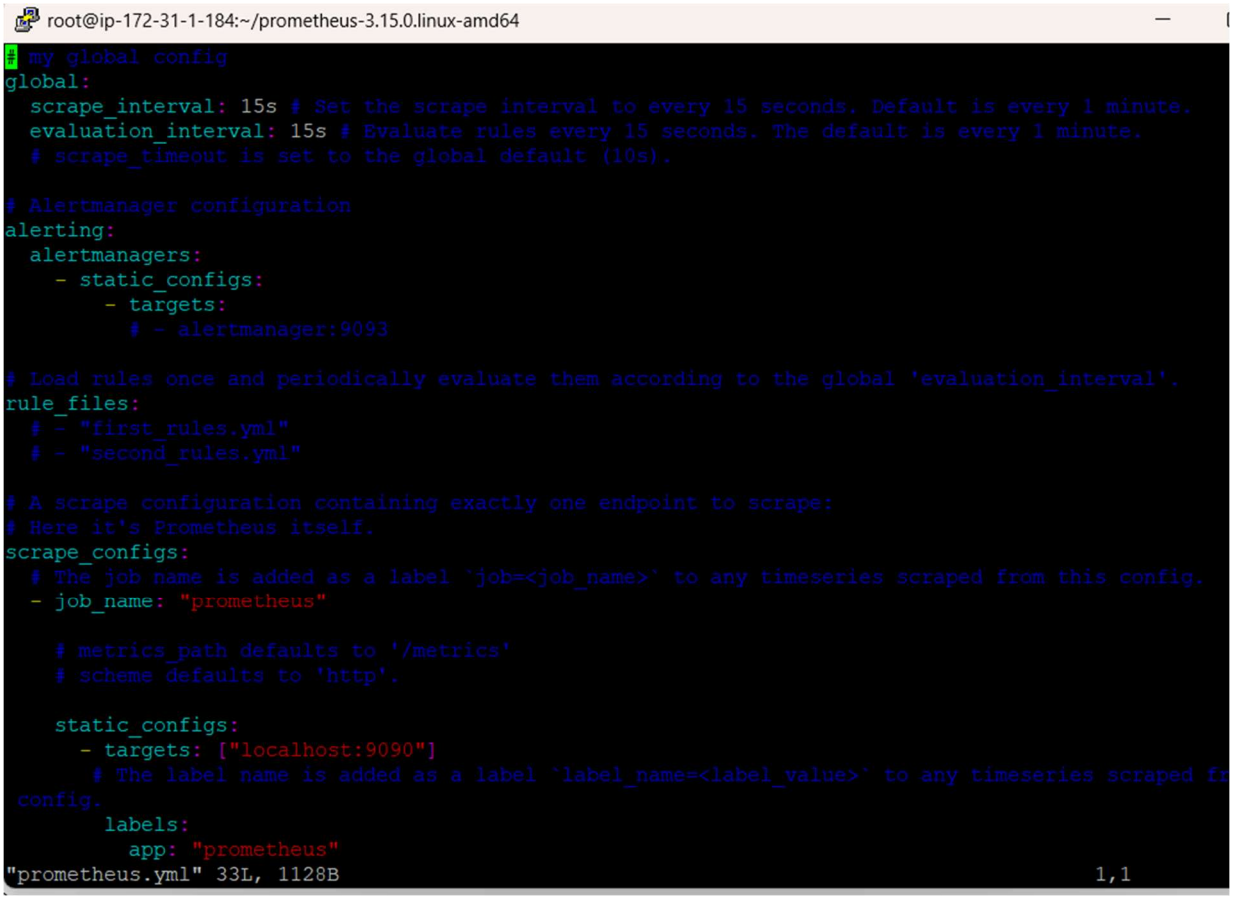
Starting the Server:



Checking the metrics



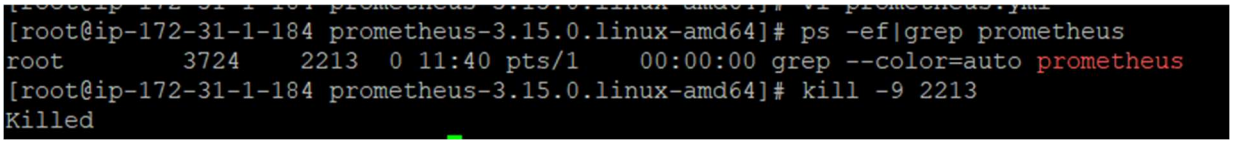
Connecting the App Server to Prometheus

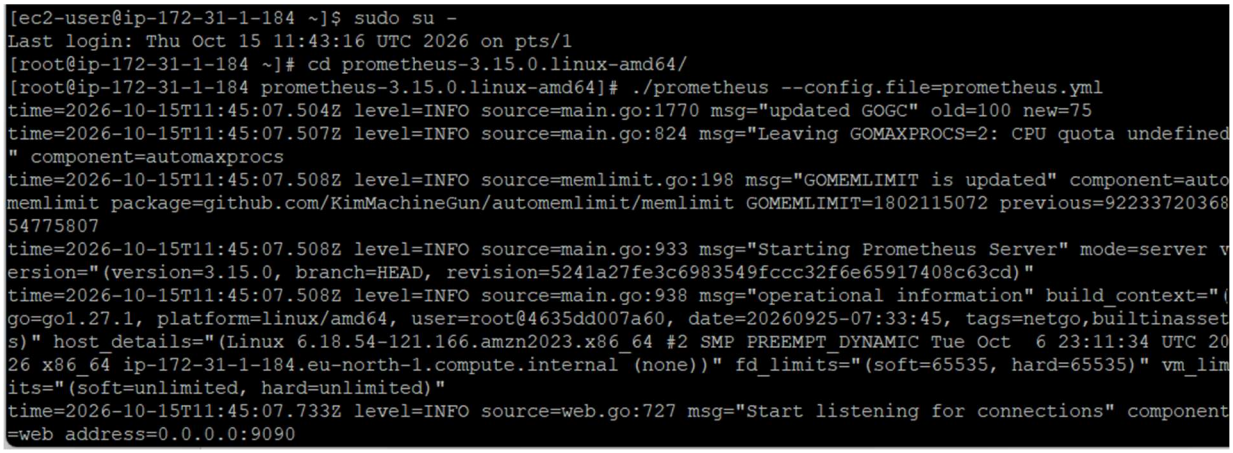


Adding the node Exporter:

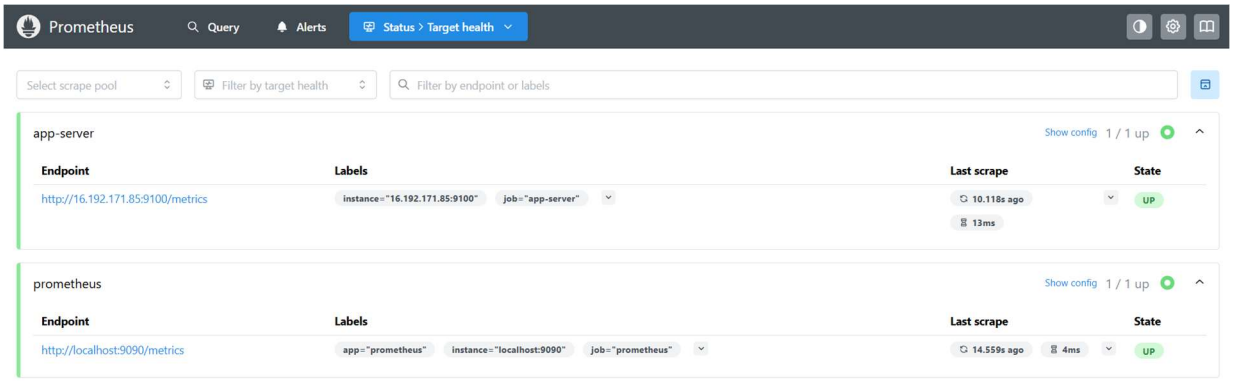


Restarting the Prometheus Server:





Checking Target Health :



Installing the Graphana :

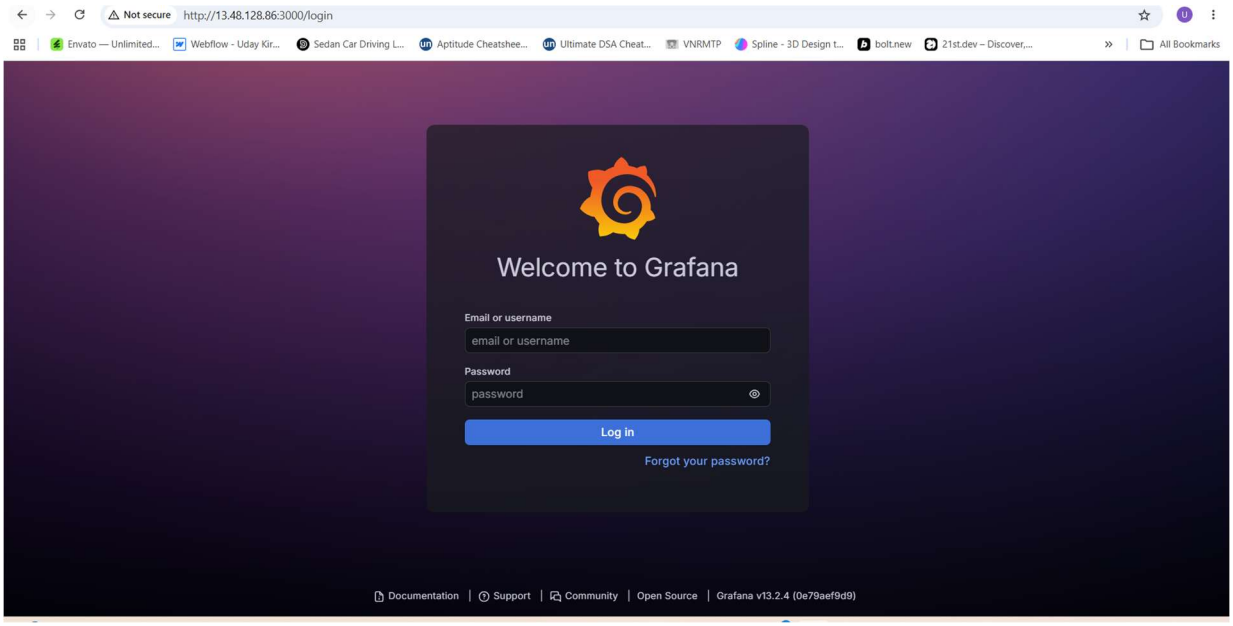
```
grafana-13.2.4/public/app/features/transformers/images/light/joinByField.svg
grafana-13.2.4/public/app/features/transformers/images/light/convertFieldType.svg
grafana-13.2.4/public/app/features/transformers/images/light/formatTime.svg
grafana-13.2.4/public/app/features/transformers/images/light/groupToNestedTable.svg
grafana-13.2.4/public/app/features/transformers/images/light/groupBy.svg
grafana-13.2.4/public/app/features/transformers/images/light/reduce.svg
grafana-13.2.4/public/app/features/transformers/images/light/partitionByValues.svg
grafana-13.2.4/public/app/features/transformers/images/light/prepareTimeSeries.svg
grafana-13.2.4/public/app/features/transformers/images/light/filterByRefId.svg
grafana-13.2.4/public/app/features/transformers/images/light/calculateField.svg
grafana-13.2.4/public/app/features/transformers/images/light/rowsToFields.svg
grafana-13.2.4/public/app/features/transformers/images/light/organize.svg
grafana-13.2.4/public/app/features/transformers/images/light/fieldLookup.svg
grafana-13.2.4/public/app/features/transformers/images/light/merge.svg
grafana-13.2.4/public/app/features/transformers/images/light/concatenate.svg
grafana-13.2.4/public/app/features/transformers/images/light/seriesToRows.svg
grafana-13.2.4/public/app/features/transformers/images/light/sortBy.svg
grafana-13.2.4/public/app/features/transformers/images/light/filterByValue.svg
grafana-13.2.4/public/app/features/transformers/images/light/formatString.svg
grafana-13.2.4/public/app/features/transformers/images/light/regression.svg
grafana-13.2.4/public/app/features/transformers/images/light/groupingToMatrix.svg
grafana-13.2.4/public/app/features/transformers/images/light/configFromData.svg
grafana-13.2.4/public/app/features/transformers/images/light/groupByCalculation.svg
grafana-13.2.4/public/app/features/transformers/images/light/joinByLabels.svg
grafana-13.2.4/public/app/features/transformers/images/light/histogram.svg
grafana-13.2.4/public/app/features/transformers/images/light/renameByRegex.svg
grafana-13.2.4/public/app/features/transformers/images/light/spatial.svg
grafana-13.2.4/public/app/features/transformers/images/light/limit.svg
```

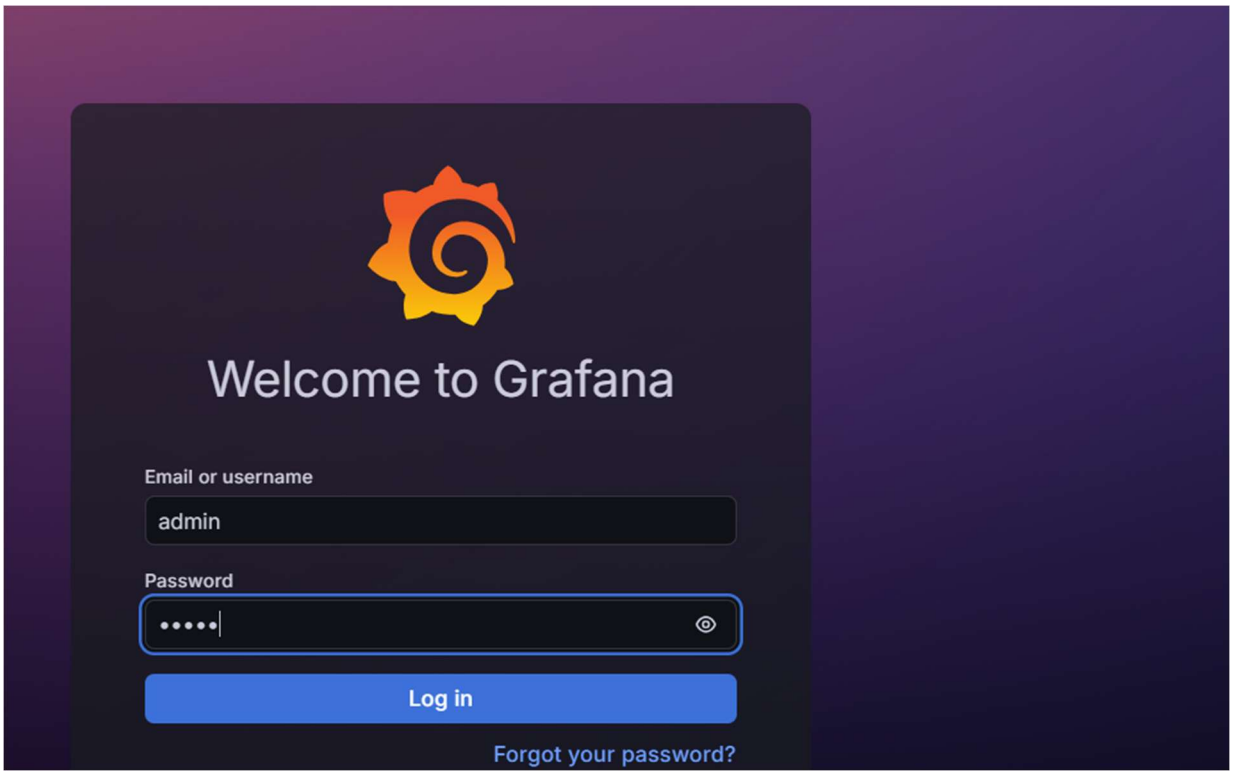
```
grafana-13.2.4/tools/zoneinfo.zip
grafana-13.2.4/plugins-bundled/
[root@ip-172-31-1-184 ~]# ls
grafana-13.2.4                                prometheus-3.15.0.linux-amd64
grafana_13.2.4_38252365904_linux_amd64.deb    prometheus-3.15.0.linux-amd64.tar.gz
grafana_13.2.4_38252365904_linux_amd64.tar.gz
[root@ip-172-31-1-184 ~]# cd grafana-13.2.4/
[root@ip-172-31-1-184 grafana-13.2.4]# ls
Dockerfile  NOTICE.md  VERSION  conf  docs      plugins-bundled  tools
LICENSE     README.md  bin      data  packaging  public
[root@ip-172-31-1-184 grafana-13.2.4]# cd bin
[root@ip-172-31-1-184 bin]#
```

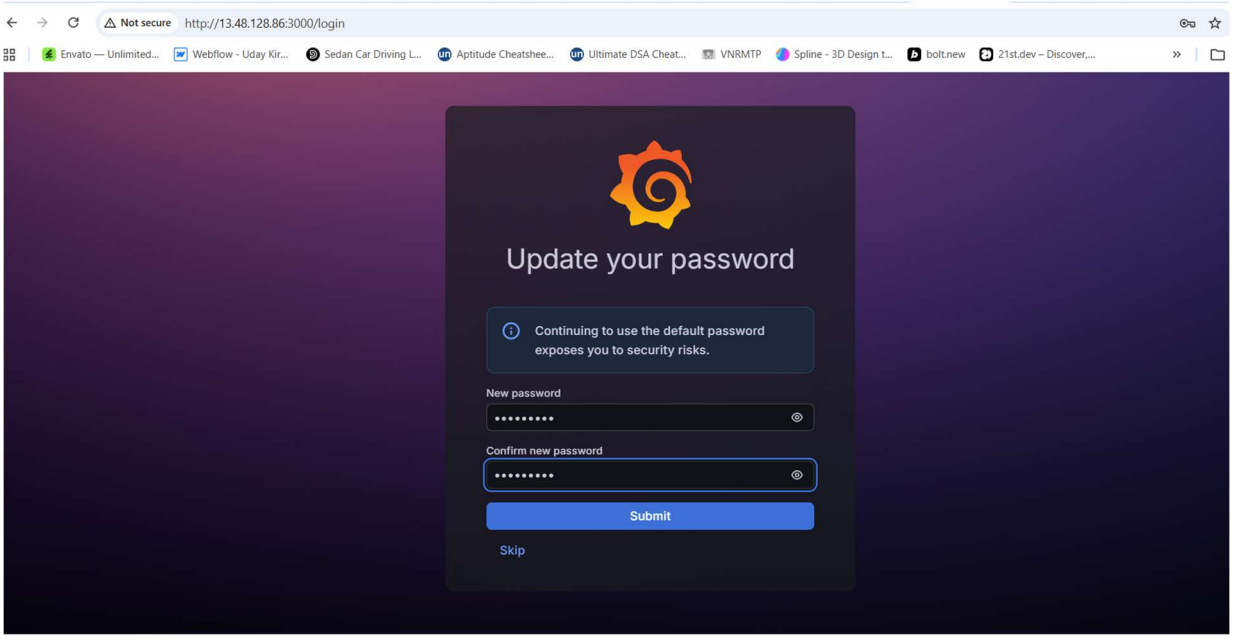
Started the Graphana

```
turls resource=shorturls.shorturl.grafana.app namespace=default legacy_count=0 unified_count=0 migrati
ummary_count=0 rejected=0 history=0
NFO [10-15|12:12:53] Migration completed for organization     logger=storage.unified.migration_runner.s
turls org_id=1 duration=11.721036ms processed=0 summaries=1 rejected=0
NFO [10-15|12:12:53] renaming legacy table                    logger=storage.unified.migrations table=s
t_url newName=short_url_legacy sql="ALTER TABLE `short_url` RENAME TO `short_url_legacy`"
NFO [10-15|12:12:53] Migration completed successfully for all organizations logger=storage.unified.migr
on_runner.shorturls org_count=1
NFO [10-15|12:12:53] Migration successfully executed          logger=unifiedstorage-migrator id="shortu
 migration" duration=18.441027ms
NFO [10-15|12:12:53] migrations completed                     logger=unifiedstorage-migrator performed=
kipped=1 duration=63.090956ms
NFO [10-15|12:12:53] Unified storage migrations completed successfully logger=storage.unified.migration
NFO [10-15|12:12:53] Live Push Gateway initialization         logger=live.push_http
NFO [10-15|12:12:53] template definitions loaded              logger=ngalert.notifier component=alertma
er orgID=1 mimir=0 grafana=0 total=0 maxTemplateOutput=10485760
NFO [10-15|12:12:53] Applying new configuration to Alertmanager logger=ngalert.notifier component=alert
ager orgID=1 configHash="{12383335334731491449 16454292769479786371 11535544170606148457 13431573875070
92 10278876130067529486 4613426124549341787 5144611013662998819}"
NFO [10-15|12:12:53] template definitions loaded              logger=ngalert.notifier component=alertma
er orgID=1 mimir=0 grafana=0 total=0 maxTemplateOutput=10485760
NFO [10-15|12:12:53] Setting up remote write using data sources logger=ngalert.writer timeout=10s defau
datasource_uid=
NFO [10-15|12:12:53] Using sync rule state persister          logger=ngalert.state.manager.persist
NFO [10-15|12:12:53] Query Service initialization             logger=query_data
NFO [10-15|12:12:53] Annotation cleanup disabled (no retention TTL configured) logger=annotation.app
```

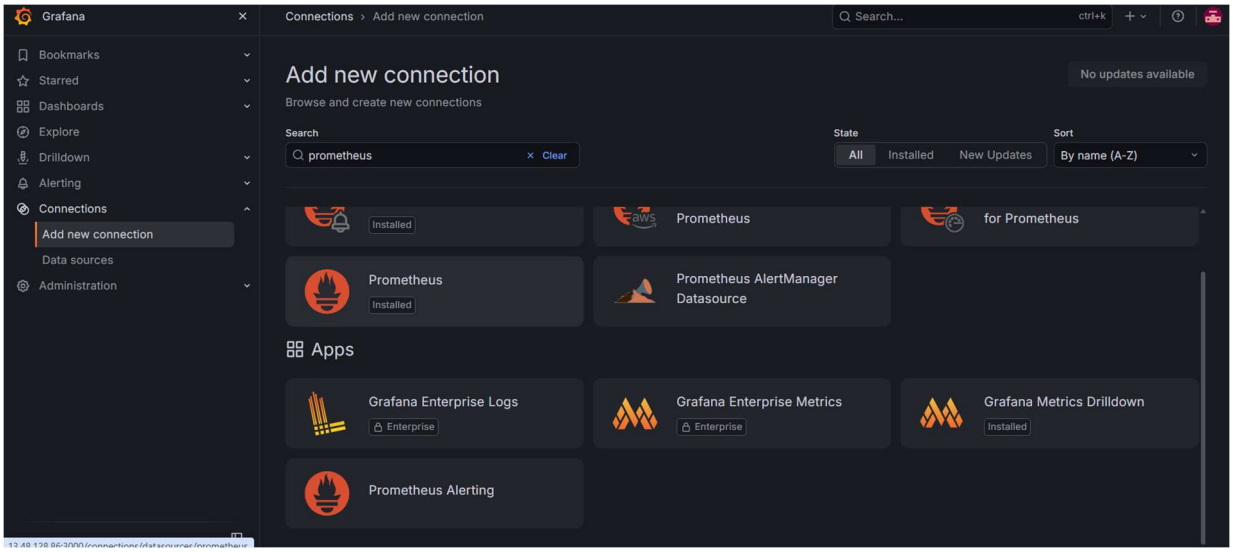
Access Graphana in browser

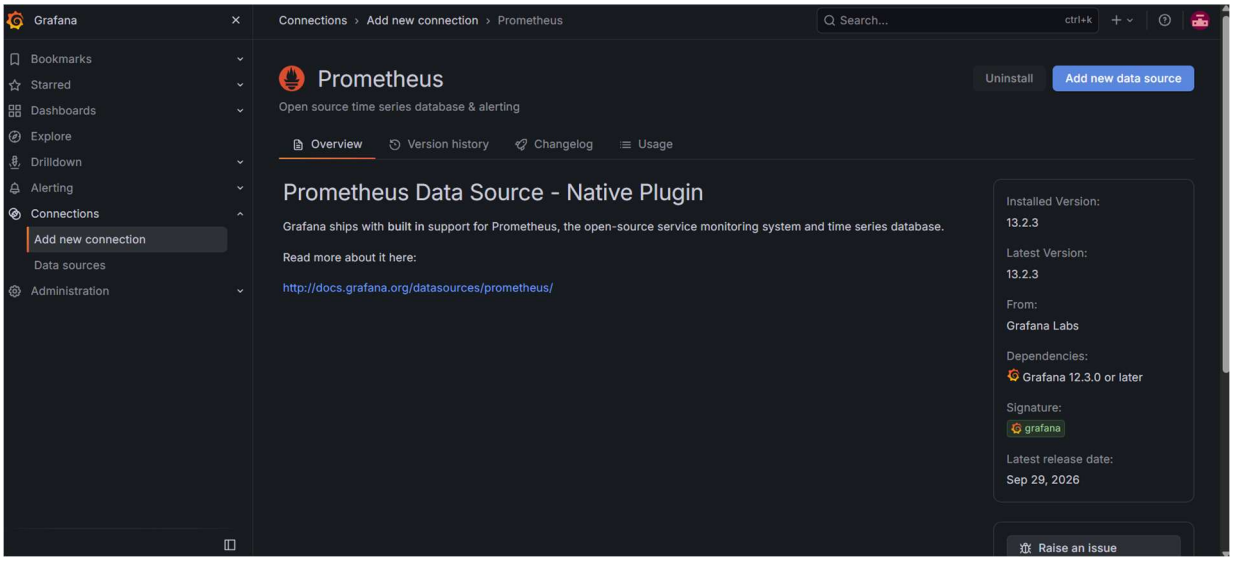


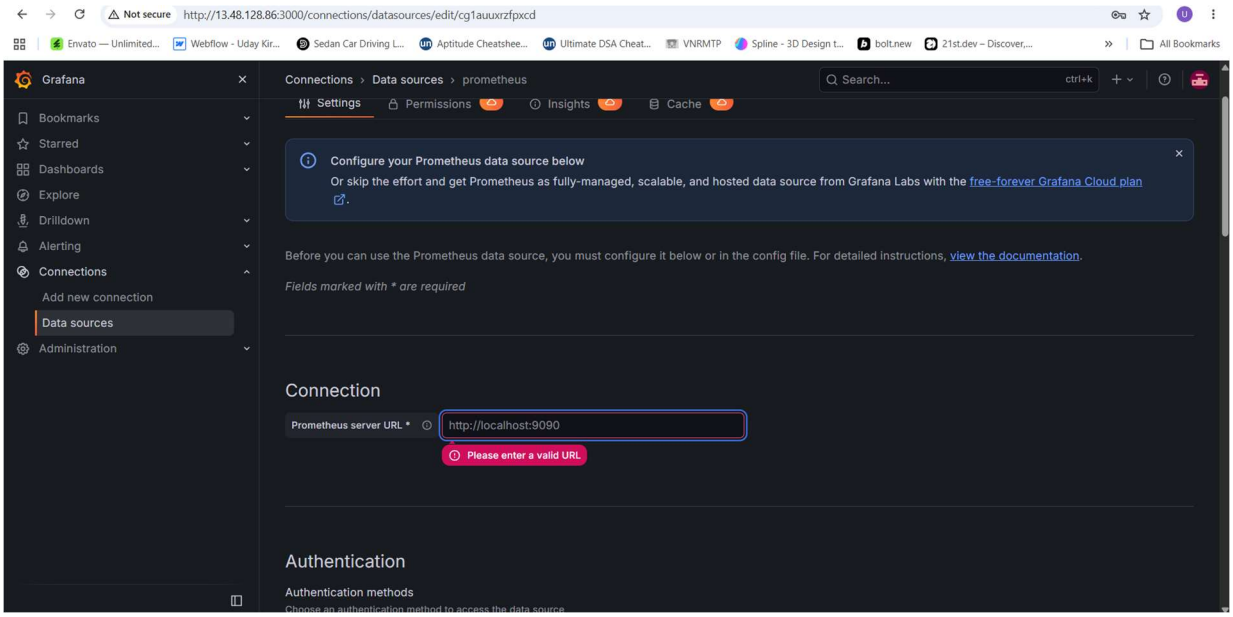

Welcome to Grafana
Email or username
admin
Password
Log in
Forgot your password?

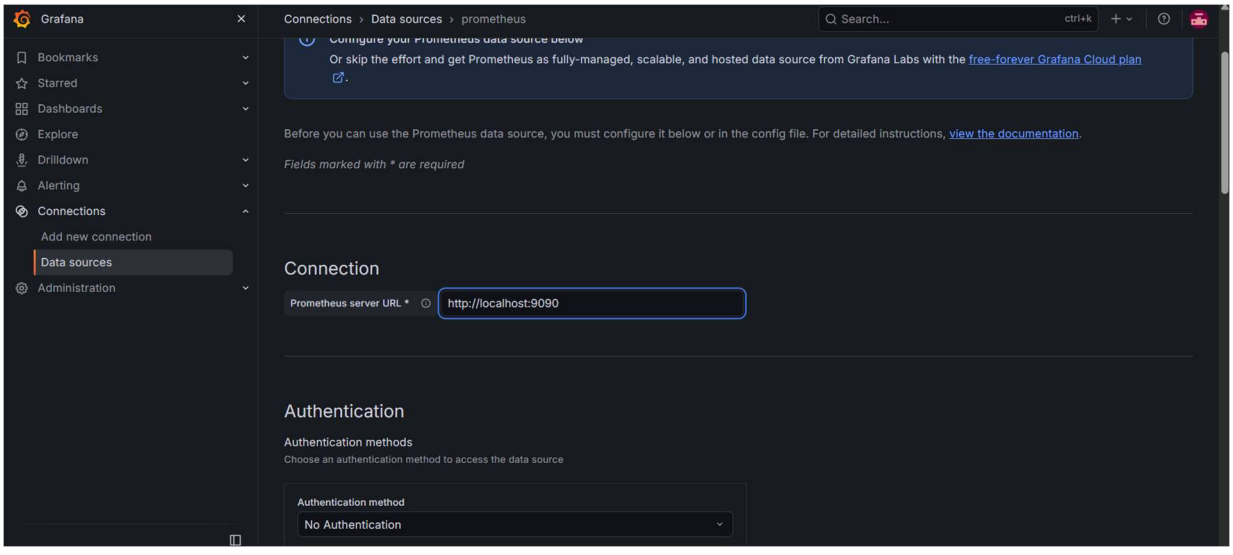

Not secure
http://13.48.128.86:3000/login
Envato — Unlimited...
Webflow - Uday Kir...
Sedan Car Driving L...
Aptitude Cheatshee...
Ultimate DSA Cheat...
VNRMTP
Spline - 3D Design t...
bolt.new
21st.dev – Discover,...
Update your password
Continuing to use the default password exposes you to security risks.
New password
Confirm new password
Submit
Skip

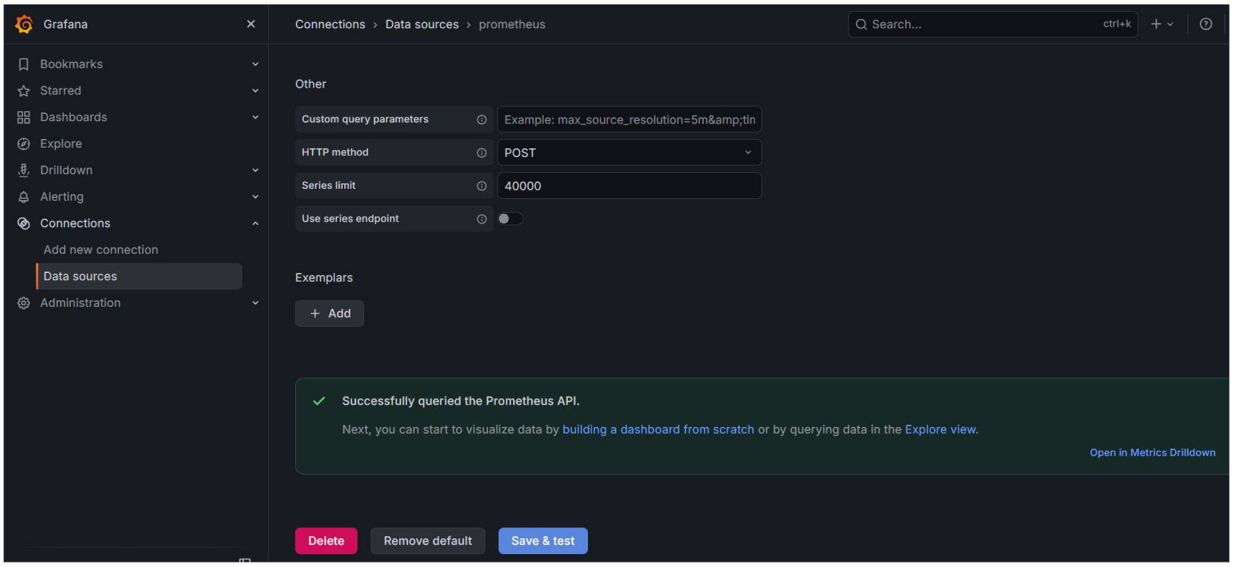
Adding the Connection :

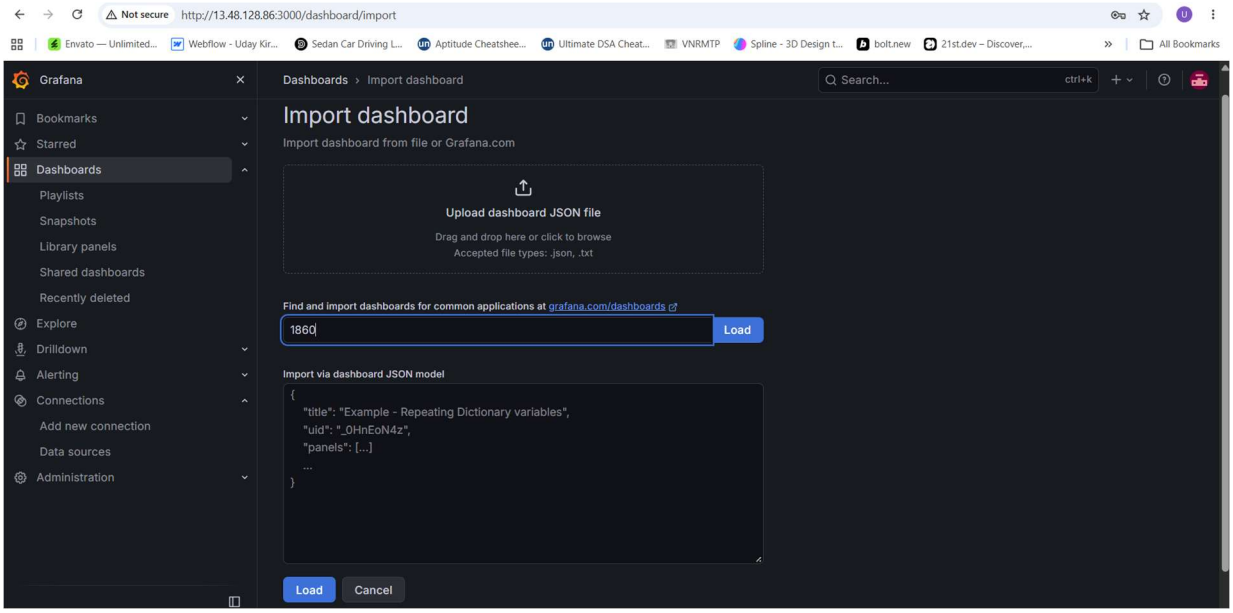




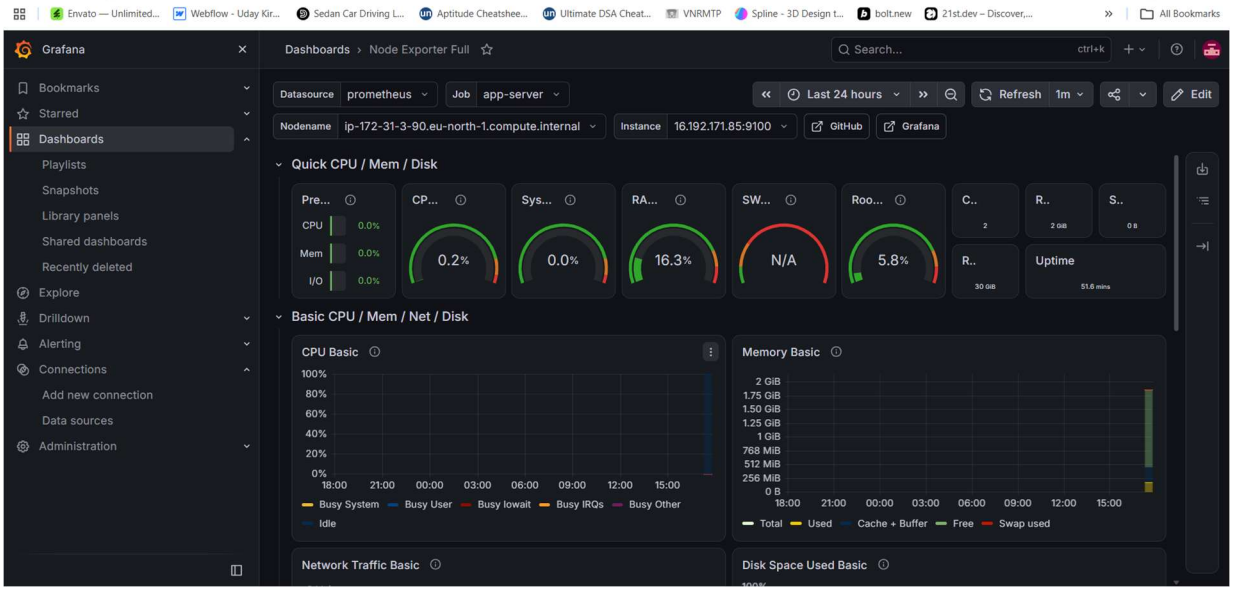

Not secure
http://13.48.128.86:3000/connections/datasources/edit/cg1auuxrzfpxcd
Grafana
Connections › Data sources › prometheus
Search...
ctrl+k
Settings
Permissions
Insights
Cache
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Configure your Prometheus data source below
Or skip the effort and get Prometheus as fully-managed, scalable, and hosted data source from Grafana Labs with the free-forever Grafana Cloud plan.
Before you can use the Prometheus data source, you must configure it below or in the config file. For detailed instructions, view the documentation.
Fields marked with * are required
Connection
Prometheus server URL *
http://localhost:9090
Please enter a valid URL
Authentication
Authentication methods

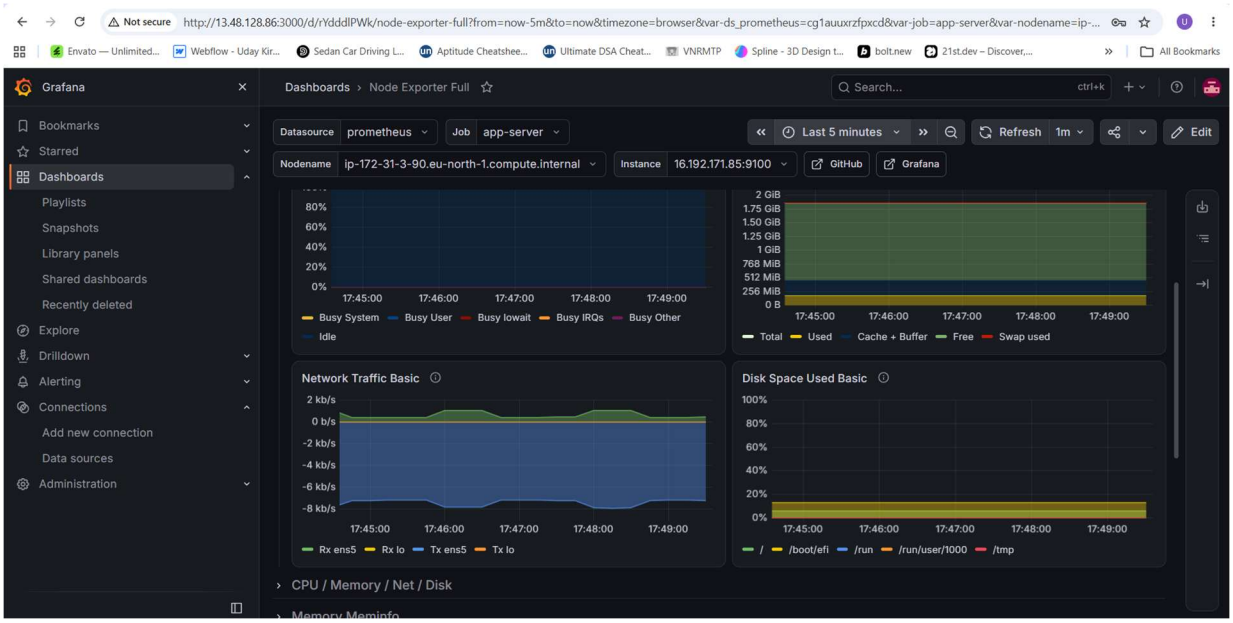

Grafana
Connections › Data sources › prometheus
Search...
ctrl+k
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Or skip the effort and get Prometheus as fully-managed, scalable, and hosted data source from Grafana Labs with the free-forever Grafana Cloud plan.
Before you can use the Prometheus data source, you must configure it below or in the config file. For detailed instructions, view the documentation.
Fields marked with * are required
Connection
Prometheus server URL *
http://localhost:9090
Authentication
Authentication methods
Choose an authentication method to access the data source
Authentication method
No Authentication

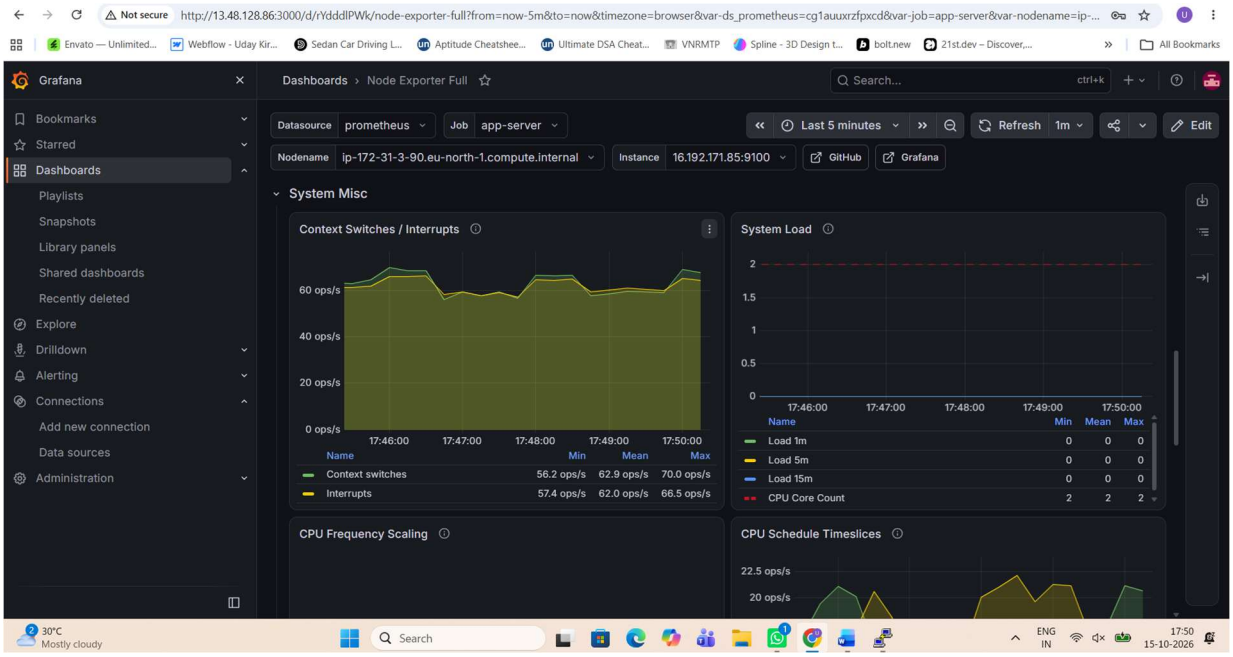




Check and access Graphana :


Grafana
Dashboards › Node Exporter Full
Search...
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Recently deleted
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Datasource prometheus
Job app-server
Last 24 hours
Refresh 1m
Edit
Nodename ip-172-31-3-90.eu-north-1.compute.internal
Instance 16.192.171.85:9100
GitHub
Grafana
Quick CPU / Mem / Disk
CPU 0.0%
Mem 0.0%
I/O 0.0%
0.2%
0.0%
16.3%
N/A
5.8%
Uptime
Basic CPU / Mem / Net / Disk
CPU Basic
Busy System
Busy User
Busy Iowait
Busy IRQs
Busy Other
Idle
Memory Basic
Total
Used
Cache + Buffer
Free
Swap used
Network Traffic Basic
Disk Space Used Basic


Not secure
http://13.48.128.86:3000/d/rYdddlPWk/node-exporter-full?from=now-5m&to=now&timezone=browser&var-ds_prometheus=cg1auuxrzfpxcd8&var-job=app-server&var-nodename=ip-...
Grafana
Dashboards › Node Exporter Full
Search...
Datasource prometheus
Job app-server
Last 5 minutes
Refresh 1m
Edit
Nodename ip-172-31-3-90.eu-north-1.compute.internal
Instance 16.192.171.85:9100
GitHub
Grafana
Busy System
Busy User
Busy Iowait
Busy IRQs
Busy Other
Idle
Total
Used
Cache + Buffer
Free
Swap used
Network Traffic Basic
Rx ens5
Rx lo
Tx ens5
Tx lo
Disk Space Used Basic
/
/boot/efi
/run
/run/user/1000
/tmp
CPU / Memory / Net / Disk


Not secure
http://13.48.128.86:3000/d/rYdddlPWk/node-exporter-full?from=now-5m&to=now&timezone=browser&var-ds_prometheus=cg1auuxrzfpxcd&var-job=app-server&var-nodename=ip-...
Grafana
Dashboards › Node Exporter Full
Search...
ctrl+k
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Recently deleted
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Datasource prometheus
Job app-server
Last 5 minutes
Refresh 1m
Edit
Nodename ip-172-31-3-90.eu-north-1.compute.internal
Instance 16.192.171.85:9100
GitHub
Grafana
System Misc
Context Switches / Interrupts
60 ops/s
40 ops/s
20 ops/s
0 ops/s
17:46:00
17:47:00
17:48:00
17:49:00
17:50:00
Name Min Mean Max
Context switches 56.2 ops/s 62.9 ops/s 70.0 ops/s
Interrupts 57.4 ops/s 62.0 ops/s 66.5 ops/s
System Load
2
1.5
1
0.5
0
17:46:00
17:47:00
17:48:00
17:49:00
17:50:00
Name Min Mean Max
Load 1m 0 0 0
Load 5m 0 0 0
Load 15m 0 0 0
CPU Core Count 2 2 2
CPU Frequency Scaling
CPU Schedule Timeslices
22.5 ops/s
20 ops/s
30°C
Mostly cloudy
Search
ENG
IN
17:50
15-10-2026